Notes: Chetty, Hendren & Katz (AER, 2016)

The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment

Contents

1 Big picture

- **Question.** What are the long-run effects of moving from high-poverty public housing projects to lower-poverty neighborhoods, especially for children who move when young?
- **Research design.** The paper reexamines the randomized Moving to Opportunity experiment using linked federal income-tax records that observe children into adulthood.
- **Main finding.** Assignment to the experimental voucher before age 13 raises individual earnings by about \$1,624 in ITT terms and \$3,477 in TOT terms for compliers.
- **Mechanism.** Better neighborhoods appear to work as childhood developmental inputs; gains decline with age at random assignment.
- **Policy message.** Mobility programs have the largest payoff when targeted to families with young children and when they induce moves to substantially better neighborhoods.

2 Data and measurement

2.1 MTO randomization and treatment arms

MTO randomized families living in high-poverty public housing in five cities into an experimental voucher group, a Section 8 voucher group, and a control group. The experimental voucher required moving to a low-poverty census tract, while the Section 8 voucher was unrestricted.

Interpretation. Random assignment creates the causal comparison; the different voucher designs create different neighborhood changes.

2.2 Linked tax data and outcomes

The paper links MTO families to federal tax records and measures adult earnings, W-2 earnings, household income, college attendance, college quality, marriage, fertility, adult zip-code characteristics, and federal income tax payments.

Interpretation. Long-run tax data are crucial because childhood exposure effects appear only after children complete schooling and enter adult labor markets.

2.3 Age split

The baseline analysis separates children below age 13 at random assignment from children age 13–18 at random assignment.

Interpretation. This split captures the exposure hypothesis: younger children spend more developmental time in the improved environment.

3 Models and empirical specifications

3.1 ITT specification

$$y_i = \alpha + \beta_E^{ITT} Exp_i + \beta_S^{ITT} S8_i + \gamma X_i + \delta_s + \varepsilon_i.$$

Interpretation. The ITT coefficients are causal effects of being offered each voucher type.

3.2 TOT / IV specification

$$y_i = \alpha_T + \beta_E^{TOT} TakeExp_i + \beta_S^{TOT} TakeS8_i + \gamma_T X_i + \delta_{Ts} + \varepsilon_{Ti}.$$

Voucher take-up is instrumented with treatment assignment.

Interpretation. TOT estimates convert voucher offers into effects of actual voucher-induced moves among compliers.

3.3 Linear exposure specification

$$y_i = \alpha + \beta_{E0} Exp_i + \beta_{S0} S8_i + \beta_{EA}(Exp_i \times AgeRA_i) + \beta_{SA}(S8_i \times AgeRA_i) + s_{a_i}\gamma + \varepsilon_i.$$

Interpretation. A negative age interaction means that treatment gains are larger for children who move earlier.

4 Identification and empirical design

The causal design relies on random assignment, strong first-stage neighborhood changes, and comparisons by age at random assignment. The main limitation is that age at move and duration of exposure are linked, so the design cannot fully separate exposure duration from age-specific disruption effects.

5 Tables and figures with expanded interpretation

5.1 Table 1: summary statistics and balance tests

Read:

- Tax-data linkage rates are high: around 86 percent for young children and 84 percent for older children.
- Young-child treatment and control groups are generally balanced on observed baseline covariates.
- Some individual covariates differ, but the overall pattern does not suggest systematic imbalance.
- The table separately reports baseline means and treatment-control differences for both age groups.

Interpretation. Table 1 is the credibility checkpoint for the randomized design in the linked tax-data sample. The important message is not that every covariate is numerically identical; it is that there is no broad pattern suggesting that treated children are systematically more advantaged before treatment.

Economic message. Table 1 is the credibility checkpoint for the randomized design in the linked tax-data sample.

What not to over-read. Balance does not prove that every subgroup estimate is precise. The older-child sample and some outcome-specific samples are smaller, so imprecision remains an issue.

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TABLE 1—SUMMARY STATISTICS AND BALANCE TESTS FOR CHILDREN IN MTO-TAX DATA LINKED SAMPLE

	< Age 13 at random assignment			Age 13–18 at random assignment		
	Control group mean (1)	Exp. versus control (2)	Sec. 8 versus control (3)	Control group mean (4)	Exp. versus control (5)	Sec. 8 versus control (6)
Linked to tax data (%)	86.4	−0.8	−0.4	83.8	1.5	−0.1
		(1.4)	(1.5)		(2.0)	(2.2)
Child's age at random assignment	8.2	−0.1	−0.0	15.1	0.1	−0.1
		(0.1)	(0.1)		(0.1)	(0.1)
Household head completed high school (%)	34.3	4.2*	0.4	29.5	5.0	0.7
		(2.4)	(2.6)		(3.1)	(3.3)
Household head employed (%)	23.8	1.0	−2.2	25.3	3.0	−0.4
		(2.1)	(2.2)		(2.9)	(3.0)
Household head gets AFDC/TANF (%)	79.5	0.6	1.8	75.0	−0.8	−1.0
		(1.9)	(2.0)		(2.9)	(3.0)
Household head never married (%)	65.1	−4.3*	−3.1	53.0	−3.1	−6.3*
		(2.3)	(2.6)		(3.2)	(3.4)
Household head had teenage birth (%)	28.6	−0.9	−0.3	29.1	−3.6	−2.5
		(2.2)	(2.5)		(2.9)	(3.2)
Primary or secondary reason for move is to get away from gangs or drugs (%)	78.1	−1.8	−4.4*	77.7	3.1	−0.9
		(2.1)	(2.4)		(2.6)	(2.9)
Household victims of crime in past five years (%)	41.3	2.5	0.9	44.8	1.3	−3.3
		(2.4)	(2.7)		(3.3)	(3.5)
Household head African American (%)	66.9	−0.4	−1.4	63.9	−1.9	−5.9**
		(2.0)	(2.1)		(2.7)	(2.8)
Household head Hispanic (%)	29.4	−0.3	−0.5		31.1	0.6
		(2.0)	(2.1)		(2.7)	(2.7)
Child susp./expelled in past two years (%)	4.9	0.7	0.4	17.6	1.0	0.4
		(0.8)	(0.9)		(2.0)	(2.2)
Children in linked MTO-tax data	1,613	1,969	1,427	686	959	686

Notes: This table presents summary statistics and balance tests for match rates and a subset of variables collected prior to randomization; online Appendix Table 1A replicates this table for all 52 control variables we use in our analysis. The estimates in the first row (fraction linked to tax data) are based on all children in the MTO data who were born in or before 1991. The estimates in the remaining rows use the subset of these observations successfully linked to the tax data. Columns 1–3 include children below age 13 at random assignment; columns 4–6 include those above age 13 at random assignment. Columns 1 and 4 show the control group mean for each variable. Columns 2 and 5 report the difference between the experimental voucher and control group, which we estimate using an OLS regression (weighted to adjust for differences in sampling probabilities across sites and over time) of each variable on indicators for being assigned to the experimental voucher group, the Section 8 voucher group, as well as indicators for randomization site. Columns 3 and 6 report the coefficient for being assigned to the Section 8 group from the same regression. The estimates in columns 2–3 and 5–6 are obtained from separate regressions. Standard errors, reported in parentheses, are clustered by family. The final row lists the number of individuals in the control, experimental, and Section 8 groups in the linked MTO-tax data sample.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

III. Analysis and Results

In our core analysis, we split the 7,340 children in our linked analysis sample

TABLE 2—FIRST-STAGE IMPACTS OF MTO ON VOUCHER TAKE-UP AND NEIGHBORHOOD POVERTY RATES (*Percentage Points*)

	Housing voucher take-up	Poverty rate in tract one year post- RA		Mean poverty rate in tract post-RA to age 18		Mean poverty rate in zip post-RA to age 18	
	(1)	ITT (2)	TOT (3)	ITT (4)	TOT (5)	ITT (6)	TOT (7)
Panel A. Children < age 13 at random assignment							
Exp. versus control	47.66***	−17.05***	−35.96***	−10.27***	−21.56***	−5.84***	−12.23***
	(1.653)	(0.853)	(1.392)	(0.650)	(1.118)	(0.425)	(0.752)
Sec. 8 versus control	65.80***	−14.88***	−22.57***	−7.97***	−12.06***	−3.43***	−5.17***
	(1.934)	(0.802)	(1.024)	(0.615)	(0.872)	(0.423)	(0.622)
Observations	5,044	4,958	4,958	5,035	5,035	5,035	5,035
Control group mean	0	50.23	50.23	41.17	41.17	31.81	31.81
Panel B. Children age 13–18 at random assignment							
Exp. versus control	40.15***	−14.00***	−34.70***	−10.04***	−24.66***	−5.51***	−13.52***
	(2.157)	(1.136)	(2.231)	(0.948)	(1.967)	(0.541)	(1.113)
Sec. 8 versus control	55.04***	−12.21***	−22.03***	−8.60***	−15.40***	−3.95***	−7.07***
	(2.537)	(1.078)	(1.738)	(0.920)	(1.530)	(0.528)	(0.921)
Observations	2,358	2,302	2,302	2,293	2,293	2,292	2,292
Control group mean	0	49.14	49.14	47.90	47.90	35.17	35.17

Notes: Columns 1, 2, 4, and 6 report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Columns 3, 5, and 7 report TOT estimates using a 2SLS specification, instrumenting for voucher take-up with the experimental and Section 8 assignment indicators. Standard errors, reported in parentheses, are clustered by family. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment The estimates in panels A and B are obtained from separate regressions. The dependent variable in column 1 is an indicator for the family taking up an MTO voucher and moving. The dependent variable in columns 2 and 3 is the census tract-level poverty rate one year after random assignment. The dependent variable in columns 4–7 is the duration-weighted mean poverty rate in the census tracts (columns 4 and 5) and zip codes (columns 6 and 7) where the child lived from random assignment till age 18. The sample in this table includes all children born before 1991 in the MTO data for whom an SSN was collected prior to RA because we were unable to link the MTO tract-level location information to the tax data. This sample is nearly identical our linked analysis sample because 99.1 percent of the children with nonmissing SSNs are matched to the tax data. The duration-weighted poverty rate is constructed using information on the addresses where the youth lived from random assignment up to their 18th birthday, weighted by the amount of time spent at each address. Census tract poverty rates in each year are interpolated using data from the 1990 and 2000 decennial censuses as well as the 2005–2009 American Community Survey, as in Sanbonmatsu et al. (2011); zip code poverty rates are from census 2000 only and are not interpolated.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.2 Table 2: first-stage impacts on voucher take-up and neighborhood poverty

Read:

- Young children: experimental assignment has a 47.66 percentage point voucher take-up rate; Section 8 has 65.80 percentage points.
- Experimental treatment reduces mean tract poverty from post-RA to age 18 by 10.27 pp ITT and 21.56 pp TOT for young children.
- Section 8 reduces the same mean tract poverty by 7.97 pp ITT and 12.06 pp TOT.
- The first-stage effects are also large for older children, though take-up is slightly lower.

Interpretation. Table 2 shows that the treatment is not merely a voucher label. It produces large and sustained differences in childhood neighborhood environments. The experimental voucher has a stronger neighborhood-quality first stage than the unrestricted Section 8 voucher because it forces or nudges moves to lower-poverty tracts.

Economic message. Table 2 shows that the treatment is not merely a voucher label.

What not to over-read. The table measures neighborhood poverty, not all neighborhood amenities. A lower-poverty tract may also differ in schools, peers, safety, transportation, and social networks, and the table does not identify which channel matters most.

5.3 Table 3: income in adulthood

Read:

- For children below 13, experimental assignment raises broad individual earnings by $1,624 ITT from 2008–2012.
- The corresponding TOT estimate is $3,476.8, roughly a 31 percent gain relative to the control mean of $11,270.
- Household income rises by $2,231 ITT for young experimental children.
- For older children, point estimates are generally negative and not statistically significant.

Interpretation. Table 3 is the paper's main economic-outcome table. The evidence says that neighborhood moves have large adult labor-market benefits for children who move early, but not for adolescents. The earnings result is especially important because earlier MTO studies found little evidence of economic effects when children were too young to observe in stable adult employment.

Economic message. Table 3 is the paper's main economic-outcome table.

What not to over-read. The estimates are early-adult earnings, not complete lifetime earnings. The paper extrapolates lifetime values later, but Table 3 itself observes only a limited part of the life cycle.

TABLE 3—IMPACTS OF MTO ON CHILDREN'S INCOME IN ADULTHOOD

	W-2 earnings ($) 2008–2012 ITT (1)	Individual earnings 2008–2012 ($) ITT (2)	ITT w/ controls (3)	TOT (4)	Individual earnings ($) Age 26 ITT (5)	2012 ITT (6)	Employed (%) 2008–2012 ITT (7)	Hhold. inc. ($) 2008–2012 ITT (8)	Inc. growth ($) 2008–2012 ITT (9)
Panel A. Children < age 13 at random assignment									
Exp. versus control	1,339.8**	1,624.0**	1,298.9**	3,476.8**	1,751.4*	1,443.8**	1.824	2,231.1***	1,309.4**
	(671.3)	(662.4)	(636.9)	(1,418.2)	(917.4)	(665.8)	(2.083)	(771.3)	(518.5)
Sec. 8 versus control	687.4	1,109.3	908.6	1,723.2	551.5	1,157.7*	1.352	1,452.4**	800.2
	(698.7)	(676.1)	(655.8)	(1051.5)	(888.1)	(690.1)	(2.294)	(735.5)	(517.0)
Observations	8,420	8,420	8,420	8,420	1,625	2,922	8,420	8,420	8,420
Control group mean	9,548.6	11,270.3	11,270.3	11,270.3	11,398.3	11,302.9	61.8	12,702.4	4,002.2
Panel B. Children age 13–18 at random assignment									
Exp. versus control	−761.2	−966.9	−879.5	−2,426.7	−539.0	−969.2	−2.173	−1,519.8	−693.6
	(870.6)	(854.3)	(817.3)	(2,154.4)	(795.4)	(1,122.2)	(2.140)	(11,02.2)	(571.6)
Sec. 8 versus control	−1,048.9	−1,132.8	−1,136.9	−2,051.1	−15.11	−869.0	−1.329	−936.7	−885.3
	(932.5)	(922.3)	(866.6)	(1,673.7)	(845.9)	(1213.3)	(2.275)	(11,85.9)	(625.2)
Observations	11,623	11,623	11,623	11,623	2,331	2,331	11,623	11,623	11,623
Control group mean	13,897.1	15,881.5	15,881.5	15,881.5	13,968.9	16,602.0	63.6	19,169.1	4,128.1

Notes: Columns 1–3 and 5–9 report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Column 4 reports TOT estimates using a 2SLS specification, instrumenting for voucher take-up with the experimental and Section 8 assignment indicators. Standard errors, reported in parentheses, are clustered by family. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. The estimates in panels A and B are obtained from separate regressions. The number of individuals is 2,922 in panel A (except in column 5, where it is 1,625) and 2,331 in panel B. The dependent variable in column 1 is individual W-2 wage earnings, summing over all available W-2 forms. Column 1 includes one observation per individual per year from 2008–2012 in which the individual is 24 or older. Column 2 replicates column 1 using individual earnings as the dependent variable. Individual earnings is defined as the sum of individual W-2 and non-W-2 earnings. Non-W-2 earnings is adjusted gross income minus own and spouse's W-2 earnings, social security and disability benefits, and UI payments, divided by the number of filers on the tax return. Non-W-2 earnings is recoded to zero if negative and is defined as zero for non-filers. Column 3 replicates column 2, controlling for the characteristics listed in online Appendix Table 1A. Column 4 reports TOT estimates corresponding to the ITT estimates in column 2. In column 5, we measure earnings in the year when the individual is 26 years old. In column 6, we measure earnings in 2012, limiting the sample to those 24 or older in 2012. Columns 7–9 replicate column 1 with the following dependent variables: employment (an indicator for having positive W-2 earnings), household income (adjusted gross income plus tax-exempt social security benefits and interest income for those who file tax returns, the sum of W-2 wage earnings, SSDI benefits, and UI benefits for non-filers, and zero for non-filers with no W-2 earnings, SSDI, or UI benefits), and individual earnings growth (the change in individual earnings between year $t-5$ and the current year t).

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

of the experimental voucher. For children aged 13–18 at RA, the estimated effects

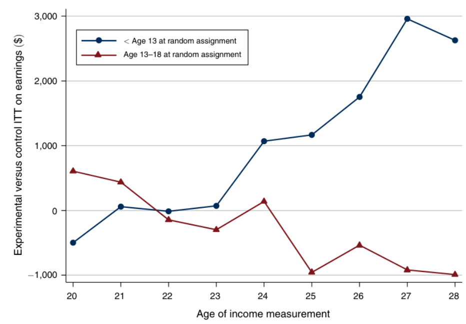

FIGURE 1. IMPACTS OF EXPERIMENTAL VOUCHER BY AGE OF EARNINGS MEASUREMENT

Notes: This figure presents ITT estimates of the impact of being assigned to the experimental voucher group on individual earnings, varying the age at which earnings is measured from 20 to 28. The estimate at each age is obtained from an OLS regression (weighted to adjust for differences in sampling probabilities across sites and over time) of individual earnings at that age on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. We plot the coefficient on the experimental voucher group indicator in this figure; the corresponding estimates for the Section 8 voucher group are shown in online Appendix Figure 1. The series in circles restricts the sample to children below age 13 at random assignment; the series in triangles includes children between age 13 and 18 at random assignment. The estimates in the two series are obtained from separate regressions. The estimates at age 26 exactly match those reported in column 5 of Table 3; the remaining estimates replicate that specification, varying the age at which earnings is measured. The null hypothesis that the experimental impacts do not vary with the age of income measurement is rejected with $p < 0.01$ for the below 13 series and $p = 0.06$ for the age 13–18 series. See notes to Table 3 for further details on specifications and variable definitions.

5.4 Figure 1: treatment effects by age of earnings measurement

Read:

- For children below age 13, the experimental-voucher effect on earnings rises as earnings are measured at older ages.
- For older children, the series is flat or slightly negative.
- The figure explains why early evaluations of MTO did not find labor-market benefits.
- Earnings impacts are small at ages 16–21 but become visible when individuals are in their mid-twenties.

Interpretation. Figure 1 is a timing diagnostic. The treatment did not immediately raise teen earnings; instead, it raised human-capital and adult-outcome trajectories that translated into earnings after schooling and early labor-market transition. This supports the paper's emphasis on long-term follow-up.

Economic message. Figure 1 is a timing diagnostic.

What not to over-read. The figure should not be read as showing that effects necessarily grow forever. It covers a young adult age range, not a full lifecycle profile.

5.5 Table 4: college attendance and college quality

Read:

- Young experimental children have a 2.509 pp higher college-attendance rate at ages 18–20.
- The same group attends colleges with average earnings quality about $686.7 higher.
- Section 8 raises college quality but not attendance as strongly.
- Older children experience negative college-attendance and college-quality effects, especially under experimental assignment.

Interpretation. Table 4 provides a likely mechanism for later earnings gains: moving young improves educational attainment and college quality. For adolescents, the negative estimates are consistent with disruption from moving during a sensitive schooling period.

Economic message. Table 4 provides a likely mechanism for later earnings gains: moving young improves educational attainment and college quality.

What not to over-read. College attendance is measured from tax forms, so it captures formal postsecondary enrollment but not all human-capital investments or informal training.

5.6 Table 5: marriage and fertility

Read:

- Young experimental children are 1.934 pp more likely to be married in adulthood.
- Young females in the experimental group are 3.341 pp more likely to be married.
- For young females, the experimental group has a 6.849 pp higher rate of father listed on birth certificate and a 4.807 pp lower rate of birth with no father present.
- For older girls, the father-on-birth-certificate effect is negative for the experimental group.

Interpretation. Table 5 shows that neighborhood exposure affects family formation, not just earnings. These outcomes are important because they indicate changes in adulthood stability and intergenerational environment, especially for daughters.

Economic message. Table 5 shows that neighborhood exposure affects family formation, not just earnings.

What not to over-read. These outcomes are measured relatively early in adulthood. Marriage and fertility patterns can evolve with age, so the table should not be read as final lifetime family-formation effects.

TABLE 4—IMPACTS OF MTO ON CHILDREN'S COLLEGE ATTENDANCE OUTCOMES

	College attendance (%) ITT					College quality ($) ITT				
	Age 18–20 (1)	Age 18 (2)	Age 19 (3)	Age 20 (4)	Age 21 (5)	Age 18–20 (6)	Age 18 (7)	Age 19 (8)	Age 20 (9)	Age 21 (10)
Panel A. Children < age 13 at random assignment										
Exp. versus control	2.509**	2.213*	2.579*	2.734*	0.409	686.7***	670.2***	800.6***	589.3**	337.8
	(1.143)	(1.200)	(1.452)	(1.464)	(1.474)	(231.2)	(240.6)	(274.3)	(262.3)	(269.9)
Sec. 8 versus control	0.992	1.221	0.502	1.252	−0.371	632.7**	592.0**	604.7**	701.4**	549.2*
	(1.264)	(1.303)	(1.613)	(1.599)	(1.592)	(256.3)	(268.2)	(304.7)	(294.9)	(293.7)
Observations	15,027	5,009	5,009	5,009	5,009	15,027	5,009	5,009	5,009	5,009
Control group mean	16.5	11.3	18.6	19.6	20.1	20,914.7	20,479.6	21,148.7	21,115.7	21,152.3
Panel B. Children age 13–18 at random assignment										
Exp. versus control	−4.261**	−5.866***	−4.460**	−2.995	−3.528*	−882.8**	−1195.7**	−890.0*	−672.6	−687.9*
	(1.712)	(2.180)	(2.162)	(2.077)	(1.972)	(385.5)	(482.8)	(465.0)	(414.2)	(402.6)
Sec. 8 versus control	−3.014*	−3.339	−3.928*	−1.882	−4.455**	−597.2	−581.5	−730.2	−492.1	−603.0
	(1.785)	(2.295)	(2.243)	(2.182)	(2.030)	(434.2)	(546.9)	(511.5)	(465.7)	(446.6)
Observations	5,100	1,328	1,722	2,050	2,234	5,100	1,328	1,722	2,050	2,234
Control group mean	15.6	12.4	16.8	16.6	17.2	21,638.0	21,337.3	21,880.1	21,629.8	21,597.8

Notes: All columns report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Standard errors, reported in parentheses, are clustered by family. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. The estimates in panels A and B are obtained from separate regressions. The dependent variable in column 1 is an indicator for attending college in a given year (having one or more 1098-T tax forms filed on one's behalf), pooling data over the three years when the individual is ages 18–20 with one observation per year per individual. Years before 1999 are excluded because 1098-T data are available beginning only in 1999. Columns 2–5 replicate column 1, using college attendance at each age between 18 and 21 as the dependent variable. The dependent variable in column 6 is Chetty, Friedman, and Rockoff's (2014) earnings-based index of college quality, again pooling data from ages 18–20 starting in 1999. This index is constructed using US population data as the mean earnings at age 31 of students enrolled in that college at age 20; children who do not attend college are assigned the mean earnings at age 31 of children who are not enrolled in any college at age 20. Columns 7–10 replicate column 6, using college quality at each age between 18 and 21 as the dependent variable.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

the repeated observations for each child. For younger children (panel A), the mean college attendance rate between the ages of 18–20 in the control group is 16.5 percent. Children assigned to the experimental voucher group are 2.5 percentage points (pp) more likely to attend college between the ages of 18–20. The corresponding TOT effect for children whose families took up the experimental voucher is a 5.2 pp

TABLE 5—IMPACTS OF MTO ON MARRIAGE AND FERTILITY (*Percentage Points*)

Sample:	All	Males	Females				
Dependent variable:	Married (1)	Married (2)	Married (3)	Has birth (4)	Teen birth (5)	Father on birth cert. (6)	Birth with no father present (7)
Panel A. Children < age 13 at random assignment							
Exp. versus control	1.934**	0.738	3.341**	−2.253	−0.670	6.849**	−4.807**
	(0.892)	(1.038)	(1.476)	(2.515)	(2.117)	(3.322)	(2.352)
Sec. 8 versus control	2.840***	1.020	4.731***	−0.285	2.409	2.671	−1.318
	(1.055)	(1.092)	(1.831)	(2.679)	(2.375)	(3.523)	(2.562)
Observations	8,420	4,384	4,036	2,409	2,379	1,410	2,409
Control group mean	3.4	2.7	4.1	59.1	19.9	44.1	33.0
Panel B. Children age 13–18 at random assignment							
Exp. versus control	−0.0637	−1.441	1.173	−2.589	−2.404	−8.259**	4.253
	(1.368)	(1.848)	(1.988)	(3.107)	(3.172)	(4.153)	(3.626)
Sec. 8 versus control	0.654	−0.577	1.811	−0.547	−0.635	−0.182	−0.701
	(1.465)	(1.946)	(2.181)	(3.304)	(3.579)	(4.409)	(3.807)
Observations	11,623	5,852	5,771	1,158	1,141	888	1,158
Control group mean	9.3	9.3	9.2	77.8	24.7	46.7	41.4

Notes: All columns report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Standard errors, reported in parentheses, are clustered by family. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. The estimates in panels A and B are obtained from separate regressions. Column 1 includes one observation per individual per year from 2008–2012 in which the individual is 24 or older. The dependent variable in column 1 is an indicator for filing a tax return as a married individual in a given year. Columns 2 and 3 replicate column 1 for males and females, respectively. Columns 4–7 restrict the sample to females and include one observation per individual. The dependent variable in column 4 is an indicator for having a child before June 2014. In column 5, it is an indicator for having a child before the age of 19. Column 6 restricts the sample to females who have a birth; the dependent variable in this column is an indicator for having a father listed on the first-born child's SSN application. In column 7, the dependent variable is an indicator for having one or more births, with no father listed on the SSN application for the first birth.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

the fraction of females who have a birth with no father present, as shown in column

5.7 Table 6: adult neighborhood characteristics

Read:

- Young experimental children live in adult zip codes with 1.592 pp lower poverty, $1,345.9 higher mean income, 2.852 pp lower black share, and 1.812 pp lower single-mother share.
- Young Section 8 children also live in better adult neighborhoods, with especially large reductions in black share and single-mother share.
- For older children, most adult-neighborhood effects are smaller and statistically weaker.

Interpretation. Table 6 shows persistence across generations: moving to a better neighborhood in childhood changes where individuals live as adults. This is a mechanism and an outcome. It may reflect better education, higher earnings, improved networks, or altered residential preferences and opportunities.

Economic message. Table 6 shows persistence across generations: moving to a better neighborhood in childhood changes where individuals live as adults.

What not to over-read. Zip-code measures are coarser than tract-level childhood measures. Adult zip-code differences should not be interpreted as perfect measures of adult neighborhood quality.

5.8 Table 7: gender heterogeneity

Read:

- For young children, benefits are generally positive for both boys and girls.
- Girls show strong improvements in marriage/family outcomes and adult neighborhood poverty.
- Boys show positive earnings effects, though standard errors are large.
- Older boys and girls generally do not show comparable positive effects.

TABLE 6—IMPACTS OF MTO ON CHILDREN'S NEIGHBORHOOD CHARACTERISTICS IN ADULTHOOD

	Poverty share in zip 2008–2012 (%) (1)	Mean income in zip 2008–2012 ($) (2)	Black share in zip 2008–2012 (%) (3)	Single mother share in zip 2008–2012 (%) (4)
Panel A. Children < age 13 at random assignment				
Exp. versus control	−1.592***	1,345.9***	−2.852**	−1.812**
	(0.602)	(489.5)	(1.417)	(0.862)
Sec. 8 versus control	−1.394**	1,322.0**	−5.654***	−3.087***
	(0.699)	(558.6)	(1.714)	(1.001)
Observations	6,649	6,649	6,651	6,648
Control group mean	23.8	25,014.3	43.0	42.0
Panel B. Children age 13–18 at random assignment				
Exp. versus control	−0.523	604.2	0.465	−0.294
	(0.643)	(478.5)	(1.654)	(0.940)
Sec. 8 versus control	−0.928	442.0	−2.631	−1.856*
	(0.698)	(524.2)	(1.715)	(0.976)
Observations	9,149	9,149	9,149	9,148
Control group mean	23.6	25,170.5	39.6	40.1

Notes: All columns report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Standard errors, reported in parentheses, are clustered by family. In this table, we only include observations where zip code information in the relevant year is available (based on 1040 tax returns, W-2's, or other information returns). In 2012, zip code data are available for 79.56 percent of observations for children age 24 or older. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. The estimates in panels A and B are obtained from separate regressions. Outcome variables are defined using zip code–level data from the 2000 census. All columns include one observation per individual per year from 2008–2012 in which the individual is 24 or older and in which zip code information is available. The dependent variable in column 1 is the poverty share (share of households below the poverty line in the 2000 census) in the individual's zip code. Columns 2–4 replicate column 1 with the following dependent variables: mean income in the zip code (aggregate income divided by the number of individuals 16–64), black share (number of people who are black alone divided by total population in 2000) and single mother share (number of households with female heads and no husband present with own children present divided by the total number of households with own children present).

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 7—HETEROGENEITY OF TREATMENT EFFECTS BY GENDER

	Experimental versus control		Section 8 versus control	
	Male (1)	Female (2)	Male (3)	Female (4)
Panel A. Children < age 13 at random assignment				
Individual earnings 2008–2012 ($) ITT	1,679.2*	1,438.9	1,073.8	1,097.6
	(973.3)	(885.1)	(947.1)	(931.6)
	[10,020.8]	[12,634.2]	[10,020.8]	[12,634.2]
College quality 18–20 ($) ITT	522.3*	819.4**	327.4	895.4**
	(290.1)	(345.0)	(290.2)	(399.2)
	[20,443.8]	[21,440.2]	[20,443.8]	[21,440.2]
Zip poverty share 2008–2012 (%) ITT	−1.498**	−1.519**	−0.733	−2.055***
	(0.720)	(0.674)	(0.793)	(0.766)
	[22.5]	[24.5]	[22.5]	[24.5]
Panel B. Children age 13–18 at random assignment				
Individual earnings 2008–2012 ($) ITT	−1,832.4	−204.9	−1,410.6	−740.6
	(1,298.3)	(1,044.6)	(1,363.6)	(1,165.4)
	[14,814.4]	[16,997.5]	[14,814.4]	[16,997.5]
College quality 18–20 ($) ITT	−590.4	−1,201.4**	−27.44	−1,105.8*
	(496.6)	(582.3)	(596.4)	(618.6)
	[20,935.2]	[22,348.0]	[20,935.2]	[22,348.0]
Zip poverty share 2008–2012 (%) ITT	0.281	−0.627	0.844	−1.241
	(1.069)	(0.954)	(1.149)	(1.059)
	[21.8]	[24.2]	[21.8]	[24.2]

Notes: This table replicates the ITT OLS regression specifications in column 2 of Table 3 (individual earnings), column 6 of Table 4 (college quality), and column 1 of Table 6 (poverty share) separately for male and female children. Columns 1 and 2 report the coefficient on the indicator for being assigned to the experimental voucher group; columns 3 and 4 report the coefficient on the indicator for being assigned to the Section 8 voucher group. The estimates in columns 1 and 3 are for males, and the two estimates in each row of these columns come from a single OLS regression analogous to that in column 2 of Table 3. The estimates in columns 2 and 4 for females are constructed analogously. Standard errors, reported in parentheses, are clustered by family. Control group means for each estimation sample are reported in square brackets. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. See the notes to Tables 3, 4, and 6 for further details on specifications and variable definitions.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

treatments also improve college quality (at ages 18–20) and neighborhood quality (at age 24 and above) for both boys and girls. Conversely, we find adverse long-term treatment effects for both boys and girls who were above age 13 at RA (panel B).

The positive effects of the MTO treatments on adult outcomes for the younger boys point to an intriguing dynamic pattern of neighborhood effects when combined

Interpretation. Table 7 asks whether the young-child result is driven only by one gender. The answer is no: the broad pattern of positive young-child effects appears for both genders, but the outcome margins differ. This strengthens the interpretation that childhood exposure has wide-ranging developmental effects.

Economic message. Table 7 asks whether the young-child result is driven only by one gender.

What not to over-read. Gender-specific estimates are less precise because the sample is split. One should focus on patterns across outcomes rather than single coefficients.

5.9 Table 8: linear exposure-effect estimates

Read:

- The experimental-voucher earnings effect at age zero is estimated at $4,823, and the effect falls by about $364 per additional year of age at RA.
- Household income, college attendance, college quality, marriage, adult neighborhood poverty, and tax payments show similar age-gradient patterns.
- The Section 8 age gradients are generally smaller or less precise, consistent with weaker neighborhood changes.

Interpretation. Table 8 gives the strongest formal evidence for exposure effects. The younger the child, the larger the predicted adult benefit. The table links the age-split results to a continuous age profile and supports the idea that years of childhood exposure matter.

Economic message. Table 8 gives the strongest formal evidence for exposure effects.

What not to over-read. The age-gradient coefficient combines exposure duration with possible age-varying disruption. The table cannot by itself isolate whether the decline is caused entirely by less exposure, more disruption at older ages, or both.

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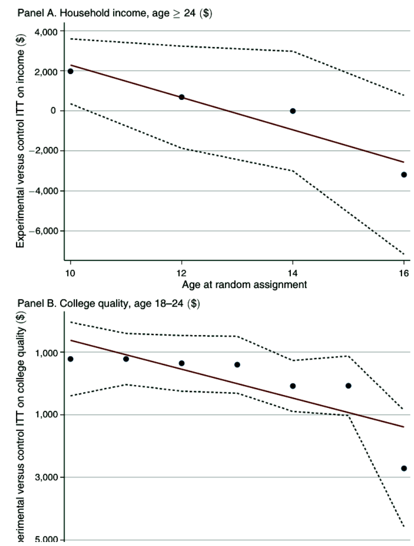
TABLE 8—LINEAR EXPOSURE EFFECT ESTIMATES

	Indiv. earn. ($) 2008–2012 ITT (1)	Household income ($) 2008–2012 ITT (2)	Household income ($) Age 26 ITT (3)	Coll. qual. 18–20 ITT ($) (4)	Married ITT (%) (5)	Zip poverty share ITT (%) (6)	Taxes paid ITT ($) (7)
Experimental × age at RA	−364.1*	−723.7***	−564.9**	−171.0***	−0.582**	0.261*	−65.81***
	(199.5)	(255.5)	(282.8)	(55.16)	(0.290)	(0.139)	(23.88)
Section 8 × age at RA	−229.5	−338.0	157.2	−117.1*	−0.433	0.0109	−42.48*
	(208.9)	(266.4)	(302.0)	(63.95)	(0.316)	(0.156)	(24.85)
Experimental	4,823.3**	9,441.1***	8,057.1**	1,951.3***	8.309**	−4.371**	831.2***
	(2,404.3)	(3,035.8)	(3,760.9)	(575.1)	(3.445)	(1.770)	(279.4)
Section 8	2,759.9	4,447.7	−1,194.0	1,461.1**	7.193*	−1.237	521.7*
	(2,506.1)	(3,111.3)	(3,868.2)	(673.6)	(3.779)	(2.021)	(287.5)
Observations	20,043	20,043	3,956	20,127	20,043	15,798	20,043
Control group mean	13,807.1	16,259.9	14,692.6	21,085.1	6.6	23.7	627.8

Notes: This table reports ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group, interactions between the experimental/Section 8 indicators and age at random assignment (RA), and interactions between randomization site indicators and age at RA. All regressions are estimated using all children in the sample with available outcome data. Standard errors, reported in parentheses, are clustered by family. The dependent variable is individual earnings in column 1 and household income in columns 2 and 3. See notes to Table 3 for definitions of these variables. In columns 4–7, we replicate column 1 with the dependent variables used in column 6 of Table 4, column 1 of Table 5, column 1 of Table 6, and column 2 of Table 12. See notes to those tables for definitions of these variables. Columns 1, 2, and 5–7 include one observation per individual per year from 2008–2012 in which the individual is 24 or older. Column 3 includes one observation per individual at age 26. Column 4 includes one observation per individual per year from ages 18–20, as in Table 4. The Experimental × age at RA coefficient can be interpreted as the change in the impact of being assigned to the experimental group for a child who is one year older at random assignment, and the Section 8 × Age at RA coefficient can be interpreted analogously.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Nonparametric Estimates by Age at Move.—In Figure 2, we evaluate how the effects of the MTO treatments vary with children's ages at move using a nonparametric approach. These figures plot ITT estimates of being assigned to the experimental voucher group by a child's age at RA, grouping children into two-year



5.10 Figure 2: impacts by age at random assignment

Read:

- The figure plots age-specific treatment effects for income, college attendance, college quality, and zip-code poverty.
- The solid fitted line slopes downward as age at random assignment increases.
- Positive effects are concentrated among younger children.
- Older-age estimates hover near zero or become negative.

Interpretation. Figure 2 visually reinforces Table 8. It shows that the age-gradient is not simply an artifact of a single arbitrary cutoff at age 13. The pattern is consistent across several outcomes.

Economic message. Figure 2 visually reinforces Table 8.

What not to over-read. Age-specific bins are noisy. The figure is best read as evidence on the broad slope, not on exact treatment effects at a single age.

5.11 Table 9: adult income outcomes

Read:

- The experimental voucher has no positive earnings effect for adults; the individual-earnings ITT is negative and insignificant.
- Section 8 also shows no meaningful positive effect on adult earnings.
- Employment and household-income impacts for adults are similarly weak.

Table 9—Impacts of MTO on Adults' Income

	Individual earnings ($)				
	2008–2012 ITT (1)	ITT w/controls (2)	2012 ITT (3)	Employed (%) ITT (4)	Hhold. inc. ($) ITT (5)
Exp. versus control	−354.1	−168.7	−681.9	−0.946	−338.5
	(621.9)	(558.2)	(675.1)	(1.704)	(740.4)
Section 8 versus control	249.5	468.1	−99.67	−0.702	516.7
	(675.2)	(609.0)	(758.4)	(1.833)	(829.4)
Observations	21,075	21,075	4,215	21,075	21,075
Control group mean	14,381.0	14,381.0	13,700.6	54.9	17,951.5

Notes: All columns report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Standard errors, reported in parentheses, are clustered by family. The sample consists of all individuals in the linked MTO-tax dataset who were not classified as children at the point of random assignment. The number of individuals is 4,215 in all columns. The dependent variable in columns 1–3 is individual earnings. Column 1 includes one observation per individual per year from 2008–2012 in which the individual is 24 or older. Column 2 replicates column 1, controlling for the predetermined characteristics listed in online Appendix Table 1A. In column 3 we replicate column 1 using data from 2012 only. Columns 4 and 5 replicate column 1 using employment and household income as the dependent variables. See notes to Table 3 for definitions of all dependent variables.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Consistent with prior work, we find no effects of MTO treatments on any of the adults' economic outcomes. The point estimates tend to be slightly negative for the experimental group and slightly positive for the Section 8 group, but all of the estimates are small and are not significantly different from zero. For example, the experimental voucher ITT on mean individual earnings from 2008–2012 is −$354 (2.4 percent of the control group mean), with a standard error of $622. The corresponding TOT estimate on individual earnings is −$734, 5.1 percent of the control group mean and 4.7 percent of the control complier mean (online Appendix Table 9B, column 1). The upper bound of the 95 percent confidence interval for the TOT estimate is $1,795, 12 percent of the control group mean. This is far below the 31 percent increase in individual earnings for young children.

Exposure Effect Estimates.—Clampet-Lundquist and Massey (2008) show that the number of years an adult spends in a low-poverty area is correlated with their earnings and other economic outcomes. Their findings raise the possibility of time

Interpretation. Table 9 is crucial because it rules out a simple story that vouchers directly improve everyone's economic outcomes. The gains appear for children exposed during development, not for adults who move after human capital is already largely formed.

Economic message. Table 9 is crucial because it rules out a simple story that vouchers directly improve everyone's economic outcomes.

What not to over-read. Adult outcomes may improve in non-economic dimensions such as health, safety, and subjective well-being, as earlier MTO work found. Table 9 is only about adult economic outcomes.

5.12 Figure 3: adult effects by years since random assignment

Read:

- The adult earnings and income treatment effects do not become positive as follow-up length increases.
- The neighborhood poverty effect falls quickly after the voucher but does not translate into adult earnings gains.
- The pattern contrasts sharply with the delayed emergence of benefits for young children.

Interpretation. Figure 3 shows that the absence of adult earnings gains is not just because impacts take time to appear. For adults, more years after random assignment do not reveal growing labor-market improvements.

Economic message. Figure 3 shows that the absence of adult earnings gains is not just because impacts take time to appear.

What not to over-read. The figure does not deny that adults benefited in health, safety, or well-being. It focuses on economic outcomes and neighborhood poverty.

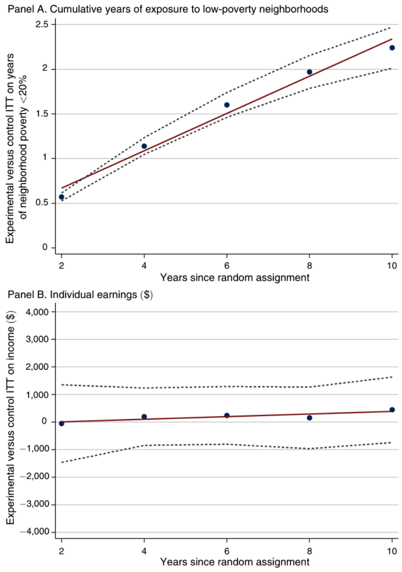


Figure 3. Impacts of Experimental Voucher on Adults by Years since Random Assignment

Notes: These figures plot ITT estimates of the impact of being assigned to the experimental voucher group by the number of years since random assignment (RA) for adults. Panel A plots impacts on the total number of years the individual lived in a census tract with a poverty rate of less than 20 percent since RA. To construct panel A, we first divide the data into two-year groups based on the number of years since RA (e.g., data in the first and second year after the calendar year of RA are assigned a value of 2). Using the data within each bin (with two observations per adult), we regress the total number of years in which the individual lived in a census tract with a poverty rate below 20 percent since RA on indicators for being assigned to the experimental and Section 8 voucher groups as well as randomization site indicators, following the standard ITT specification used for other outcomes. The solid line is a best fit line for the plotted estimates. Tract poverty rates were linearly interpolated using data from the 1990 and 2000 decennial censuses as well as the 2005–2009 American Community Survey. Panel B plots impacts on individual earnings, and is constructed using the same approach as in panel A. The regression specification used to estimate the coefficients plotted in panel B is analogous to that in column 1 of Table 9, with one observation per adult at age

We first verify that the total time of exposure to low-poverty environments increases with time since RA for adults who were assigned to the experimental voucher group relative to the control group. Prior studies have observed that some MTO participants in the experimental group moved back to higher-poverty areas over time, while some families in the control group moved to lower-poverty areas over time (e.g., Clampet-Lundquist and Massey 2008). We assess the impacts of such subsequent moves in panel A of Figure 3. We regress the cumulative number of years that the adult lived in a census tract with a poverty rate below 20 percent since RA on the MTO treatment indicators. The figure plots the ITT effects of the experimental voucher on cumulative exposure to low-poverty areas versus the number of years since RA. It is clear that the total amount of exposure to low-poverty areas rises substantially over time in the experimental group relative to the control group despite the fact that some families moved again in subsequent years.

Panel B of Figure 3 shows ITT effects of the experimental voucher on adults' individual earnings by years since RA, estimated using regressions analogous to that in column 1 of Table 9. The estimated impact on adult earnings is consistently close to zero when measuring earnings in the one to ten years after RA, with no evidence of the increasing pattern that one would expect if time of exposure in adulthood has a causal effect. The results are very similar for the Section 8 treatment (online Appendix Figure 3). We conclude that exposure to improved neighborhood environments—at least for the range of moves generated by the MTO experiment—has little impact on adults' economic outcomes.[40]

Together with our findings in Section III, the results in Figure 3 show that it is the amount of exposure to better neighborhoods *during childhood* (rather than total lifetime exposure) that matters for long-term economic success. Moreover, these findings imply that the MTO treatment effects on children's outcomes do not arise from improvements in family income. Instead, they are likely to be driven by direct effects of neighborhood environments on the children or to be mediated by parental health and stress, which were improved by the MTO treatments (Ludwig et al. 2011, Ludwig et al. 2012).

5.13 Figure 3 continued: neighborhood exposure for adults

Read:

- The continuation of Figure 3 emphasizes that adult neighborhood effects fade or stabilize without creating rising earnings effects.
- The contrast helps separate child-exposure mechanisms from contemporaneous neighborhood effects on adult workers.

Interpretation. The figure supports the lifecycle interpretation: neighborhoods are more powerful when they affect children's development than when they affect adults' current labor-market opportunities.

Economic message. The figure supports the lifecycle interpretation: neighborhoods are more powerful when they affect children's development than when they affect adults' current labor-market opportunities.

What not to over-read. The adult null result may not generalize to every adult population or every type of neighborhood intervention.

5.14 Table 10: comparison with prior MTO final evaluation

Read:

- Column 1 replicates the young-child earnings effect using 2008–2012 data: \$1,624.
- Pooling all children hides the effect: column 2 gives only \$302.5.
- Using only 2008 data reduces power and changes the sample.
- Measuring earnings at ages 16–21 shows essentially no effect, matching why earlier studies did not detect the long-run gains.

Interpretation. Table 10 reconciles this paper with earlier MTO evaluations. The new result does not contradict prior findings; rather, it uses better timing and a more theoretically motivated subgroup. The earnings gains appear after education and early labor-market transition.

Economic message. Table 10 reconciles this paper with earlier MTO evaluations.

What not to over-read. The comparison should not be read as saying prior work was wrong. Prior work answered a different question with shorter follow-up and less mature child cohorts.

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TABLE 10—MTO IMPACTS ON CHILDREN'S EARNINGS: COMPARISON TO MTO FINAL IMPACTS EVALUATION

Dependent variable:	Individual earnings				Individual earnings measured age 16–21 (\$)	
	Measured age $\geq$ 24 (\$)					
Sample:	< Age 13 2008–2012 (1)	All children 2008–2012 (2)	< Age 13 2008 (3)	All children 2008 (4)	< Age 13 Up to 2012 (5)	All children 2008 (6)
Exp. versus control	1,624.0**	302.5	1,840.9	−236.6	−30.97	−286.3
	(662.4)	(578.2)	(1339.7)	(757.0)	(229.7)	(410.8)
Section 8 versus control	1,109.3	−44.06	2,860.1*	−213.7	197.4	190.0
	(676.1)	(621.5)	(1,486.1)	(799.9)	(176.6)	(351.8)
Observations	8,420	20,043	552	2,851	30,011	3,384
Control group mean	11,270.3	13,807.1	11,615.0	14,531.5	4,033.3	4,923.0

Notes: All columns report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Standard errors, reported in parentheses, are clustered by family. The dependent variable is individual earnings in all columns, defined in the notes to Table 3. Column 1 replicates the specification in column 2 of Table 3, panel A, and includes children below age 13 at age of random assignment. This specification includes one observation per individual per year from 2008–2012 in which the individual is 24 or older. Column 2 replicates column 1, pooling all children (irrespective of age at random assignment) in the sample. Columns 3 and 4 replicate columns 1 and 2, limiting the sample to data from the 2008 tax year, which was the last year of data available for the MTO final impacts evaluation (Sanbonmatsu et al. 2011). Columns 3 and 4 therefore only include children who were 24 or older in 2008. Column 5 replicates column 1, with one observation per year in which the child is between the ages of 16–21, using all years in which we observe individual earnings (1999–2012). Column 6 replicates column 4, restricting the sample to children between the ages of 16 and 21 in 2008, as in the MTO final impacts evaluation.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

under 13 at RA). The remaining columns present variants of this specification that highlight three reasons why our findings differ from prior results.

First, if we had followed earlier MTO work in pooling younger and older MTO children, we also would have found no mean effects on earnings in adulthood, as shown in column 2 of Table 10. Such pooled estimates hide the positive MTO effects on younger children and negative effects on older children.

Second, we measure the earnings of children who were 24 years or older in 2012 in our data. If instead we had conducted our analysis in 2008—the time of the MTO final impacts evaluation—we would have found positive but very imprecisely estimated effects on earnings for children who were below age 13 at RA, as shown in column 3. This is because one would have had only 552 observations on earnings for children who were less than 13 at RA in 2008. If one had attempted to expand the sample by including all children in the analysis, as in column 4, one would have

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TABLE 11—MULTIPLE COMPARISONS: F-TESTS FOR SUBGROUP HETEROGENEITY

	Individual earnings 2008–2012 (\$) (1)	Household income 2008–2012 (\$) (2)	College attendance 18–20 (%) (3)	College quality 18–20 (\$) (4)	Married (%) (5)	Poverty share in zip 2008–2012 (%) (6)
Panel A. p-values for comparisons by age group						
Exp. versus control	0.0203	0.0034	0.0035	0.0006	0.0814	0.0265
Section 8 versus control	0.0864	0.0700	0.1517	0.0115	0.0197	0.0742
Exp. and Section 8 versus control	0.0646	0.0161	0.0218	0.0020	0.0434	0.0627
Panel B. p-values for comparisons by age, site, gender, and race groups						
Exp. versus control	0.1121	0.0086	0.0167	0.0210	0.2788	0.0170
Section 8 versus control	0.0718	0.1891	0.1995	0.0223	0.1329	0.0136
Exp. and Section 8 versus control	0.1802	0.0446	0.0328	0.0202	0.1987	0.0016

Notes: This table presents p-values for nonzero MTO treatment effects in subgroups for selected outcomes analyzed in Tables 3 to 6. In panel A, we regress the outcome on the MTO treatment indicators (*Exp* and *S*8) interacted with an indicator for being below age 13 at RA (Below13), including site dummies as controls, and clustering standard errors by family. We then run F-tests for the null hypothesis that the Exp and Exp-Below13 interaction effect are both 0 (row 1), the *S*8 and *S*8-Below13 interaction effect are both 0 (row 2), and both sets of treatment effect estimates are 0 (row 3). In panel B, we regress the outcomes on the MTO treatment indicators interacted with the following subgroup indicators: the five randomization sites, racial groups (black, Hispanic, and other), gender, and age at RA below 13. As in panel A, we then test the hypothesis that the experimental indicator and all of its subgroup interactions are 0 (row 1), the Section 8 indicator and all of its interactions are 0 (row 2), and both sets of treatment effects are 0 (row 3). See notes to Tables 3–6 for definitions of the outcome variables.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

have the most precise estimates in our baseline analysis—e.g., the college outcomes and household income—we reject the null of zero treatment effects in all subgroups with $p < 0.05$ both for the experimental versus control comparison in row 1 and the pooled comparison in row 3.

As an alternative to the parametric F-test, we implement a nonparametric permutation test for subgroup heterogeneity using the p-values from our OLS regressions as critical values, as in Ding, Feller, and Miratrix (2015). We generate 5,000 "placebo" samples in which we randomly reassign treatment status to families within randomization sites. In each placebo sample, we estimate the experimental and

5.15 Table 11: multiple comparisons

Read:

- The age-group heterogeneity tests reject zero effects for several outcomes, especially college outcomes and household income.
- The broader age/site/gender/race subgroup tests also find significant patterns for many outcomes.
- The table addresses the concern that the below-age-13 result could be a chance subgroup discovery.

Interpretation. Table 11 is a statistical credibility table. It shows that the young-child pattern is unlikely to arise simply because the authors searched many subgroups. This is especially persuasive because the age-exposure hypothesis was motivated by prior evidence.

Economic message. Table 11 is a statistical credibility table.

What not to over-read. Multiple-comparison corrections reduce but do not eliminate all judgment. The economic coherence of the pattern across earnings, education, neighborhood, and family outcomes remains important.

5.16 Table 12: federal income tax payments

TABLE 12—IMPACTS OF MTO ON FEDERAL INCOME TAX PAYMENTS

	Filing a tax return (%) 2008–2012 ITT (1)	Total income taxes paid ($) 2008–2012 ITT (2)	Total income taxes paid ($) 2008–2012 TOT (3)
Panel A. Children < age 13 at random assignment			
Exp. versus control	5.748***	183.9***	393.6***
	(2.055)	(62.80)	(134.1)
Section 8 versus control	4.789**	109.0**	169.1**
	(2.237)	(54.76)	(85.48)
Observations	8,420	8,420	8,420
Control group mean	59.3	447.5	447.5
Panel B. Children age 13–18 at random assignment			
Exp. versus control	−2.079	−175.9*	−441.6*
	(2.055)	(91.15)	(230.8)
Section 8 versus control	−2.248	−127.1	−230.1
	(2.249)	(95.52)	(173.2)
Observations	11,623	11,623	11,623
Control group mean	65.6	775.2	775.2

Notes: Columns 1 and 2 report ITT estimates from OLS regressions (weighted to adjust for differences in sampling probabilities across sites and over time) of an outcome on indicators for being assigned to the experimental voucher group and the Section 8 voucher group as well as randomization site indicators. Column 3 reports TOT estimates using a 2SLS specification, instrumenting for voucher take-up with the experimental and Section 8 assignment indicators. Standard errors, reported in parentheses, are clustered by family. Panel A restricts the sample to children below age 13 at random assignment; panel B includes children between age 13 and 18 at random assignment. The estimates in panels A and B are obtained from separate regressions. The dependent variable in column 1 is an indicator for filing a 1040 tax return. In columns 2 and 3, the dependent variable is total taxes paid, defined as the total tax field from form 1040 for filers and total taxes withheld on W-2 forms for non-filers. Columns 1 and 2 include one observation per individual per year from 2008–2012 in which the individual is 24 or older. Column 3 reports TOT estimates corresponding to the ITT estimates in column 3.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

families with young children out of high-poverty housing projects will most likely save the government money.

Policy Implications.—We now return to the policy questions posed at the beginning of this section. On the narrower question of comparing MTO-type experimental vouchers to project-based public housing, the data strongly suggest that vouchers targeted at families with young children are likely to yield net gains. Indeed, such a

5.16 Table 12: federal income tax payments

Read:

- Young experimental children are 5.748 pp more likely to file a tax return.
- They pay $183.9 more in federal income taxes in ITT terms and $393.6 more in TOT terms.
- Section 8 produces smaller but positive tax-payment effects for young children.
- Older children show negative or weak tax-payment impacts.

Interpretation. Table 12 connects the treatment to public finance. Higher adult earnings among young movers feed back into higher tax revenue, partially offsetting the cost of vouchers. This is central for the paper's policy argument.

Economic message. Table 12 connects the treatment to public finance.

What not to over-read. Tax payments are measured in early adulthood only. The fiscal return over the full lifecycle depends on the persistence of earnings gains and on later tax behavior.

6 Short summary

- MTO randomly offered vouchers to families in high-poverty public housing.
- Experimental vouchers induced moves to substantially lower-poverty areas.
- Young children gained in earnings, college attendance, college quality, family outcomes, adult neighborhoods, and tax payments.
- Older children did not benefit and sometimes experienced negative effects.
- Effects decline with age at random assignment, supporting a childhood exposure interpretation.
- Adults did not experience comparable earnings gains.
- Prior MTO evaluations missed these gains because they pooled ages and observed children too early.

7 Expanded conclusion

7.1 Core conclusion

Exposure to better neighborhoods during childhood has large long-run effects, especially for children who move before adolescence.

7.2 Economic intuition

Neighborhoods work like developmental inputs. Better schools, peers, safety, networks, and norms matter most when children are exposed for many years.

7.3 Policy implication

Housing mobility policy should target families with young children and help them move to genuinely higher-opportunity neighborhoods, not merely provide unrestricted rental assistance.

7.4 Fiscal implication

Higher adult earnings raise future tax payments, partly offsetting voucher costs.

7.5 Limits

The experiment cannot fully decompose exposure duration from disruption costs, and the findings apply most directly to families starting in high-poverty public housing.

7.6 Final takeaway

Place matters, but timing matters more. Moving early can change adult trajectories; moving later is not the same intervention.